

Are Multimodal Large Language Models Ready for Omnidirectional Spatial Reasoning?

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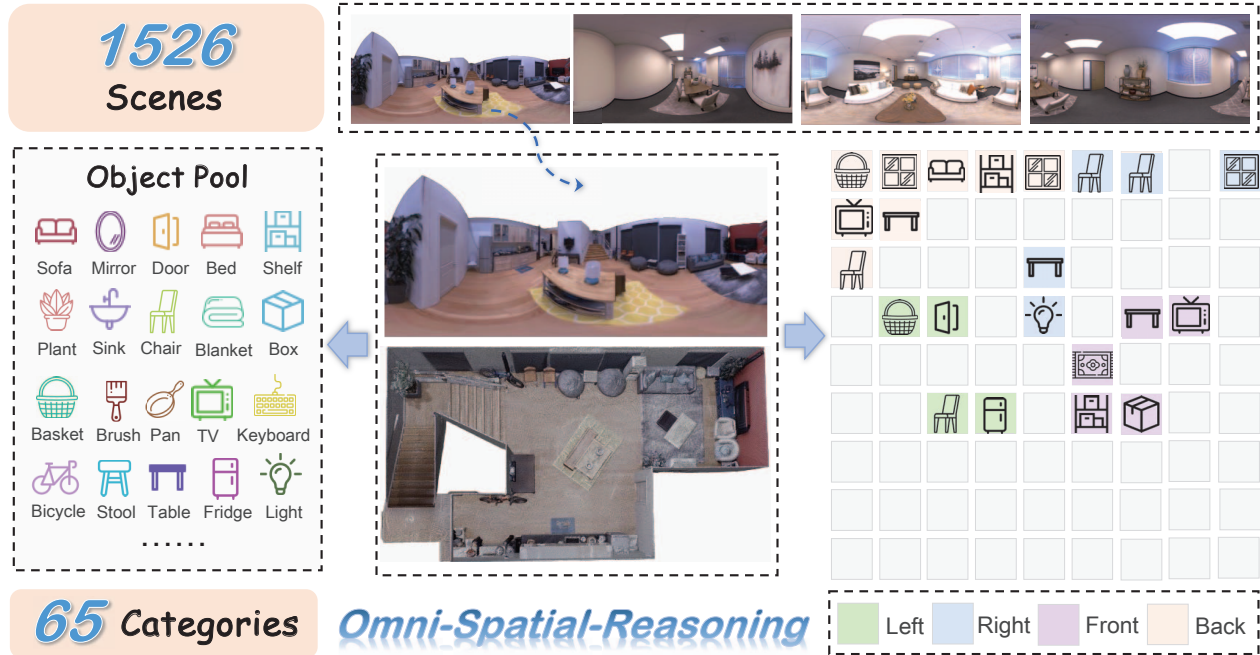


Figure 1. Overview of our *Omni-Spatial-Reasoning (OSR)* dataset and benchmark.

Abstract

The $180^\circ \times 360^\circ$ omnidirectional field of view captured by 360° cameras facilitates their application across a broad spectrum of downstream tasks such as embodied AI and virtual reality. Despite recent progress in exploring the visual-spatial reasoning capabilities of Multimodal Large Language Models (MLLMs), most existing efforts focus on pinhole-view images, leaving omnidirectional perception underexplored. In this paper, we ask: Are MLLMs ready for omnidirectional spatial reasoning? To address this, we introduce **OSR-Bench**, the first benchmark specifically de-

signed for omnidirectional spatial reasoning. *OSR-Bench* comprises over 153K diverse Question-Answer (QA) pairs grounded in high-fidelity omni-cognitive maps extracted from panoramic indoor scenes, covering key reasoning categories including object counting, relative distance, and relative direction. We further propose a negative sampling strategy that injects non-existent objects into prompts to systematically evaluate hallucination and grounding robustness. To support fine-grained analysis, we design a two-stage evaluation framework that assesses both cognitive map generation and QA accuracy using rotation-invariant matching and both rule-based and LLM-based metrics. We benchmark eight state-of-the-art MLLMs, including GPT-4o,

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Gemini 1.5 Pro, and open-source counterparts, under zero-shot settings. Our results reveal that current models exhibit limited spatial reasoning capabilities under panoramic settings, highlighting the need for more robust and perceptually grounded MLLMs. The established OSR-Bench and source code are publicly available at <https://huggingface.co/datasets/UUUuserna/OSR-Bench>.

1. Introduction

Omnidirectional images, captured using 360° cameras, offer a full 180°×360° field of view, enabling comprehensive perception of surrounding environments. This panoramic imaging capability has driven the adoption of omnidirectional sensors in numerous downstream applications, including Augmented Reality (AR), Virtual Reality (VR), and scene understanding tasks [15, 16, 21, 29, 54, 55]. Unlike conventional pinhole cameras that provide a limited field of view, 360° cameras capture spatial context in their entirety, making them especially advantageous for embodied AI and spatial reasoning scenarios. Consequently, there has been increasing research interest in leveraging omnidirectional perception for fine-grained scene understanding, particularly in tasks such as semantic segmentation [4, 44–47, 49, 51–53, 56, 57].

In parallel, the emergence of Large Language Models (LLMs) has brought transformative advances in multi-step reasoning, with growing efforts to extend these capabilities into the visual domain through Multimodal Large Language Models (MLLMs) [14, 19, 36, 39]. While MLLMs have demonstrated remarkable performance across modalities such as vision, text, and audio, most existing approaches are built upon 2D planar images from pinhole cameras, thereby under-utilizing the rich spatial information encoded in omnidirectional views. To bridge this gap, we present **OSR-Bench**—the first benchmark specifically designed to evaluate the spatial reasoning abilities of MLLMs in 360° panoramic environments. Built upon the ReplicaPano [10] and DeepPanoContext [43] datasets, OSR-Bench leverages high-fidelity 3D layout annotations to construct ground-truth top-down *omni-cognitive maps*, which are used both as evaluation references and as structured priors for generating diverse spatial reasoning questions, as shown in Figure 1.

We develop a scalable, templated QA generation pipeline that produces question-answer pairs covering three key spatial reasoning categories: object counting, relative distance, and relative direction, with cases shown in Figure 2. Each QA pair is derived from the panoramic image and its 3D layout, enabling direct grounding in visual geometry. To assess the robustness of visual understanding, we further introduce a *negative sampling strategy* inspired by recent hallucination detection techniques [20]. By injecting non-existent distractor objects into prompts and QA templates, we generate both matched and mismatched question variants

to systematically probe visual grounding fidelity. Our final benchmark comprises over 153,000 QA pairs across 4,100 panoramic images from 1,526 indoor scenes and 65 object categories. OSR-Bench supports fine-grained, interpretable evaluation under both clean and adversarial conditions, offering a rigorous testbed for analyzing the spatial reasoning and robustness of MLLMs in realistic, panoramic settings.

To enable systematic and reproducible evaluation, we design a two-stage assessment protocol tailored for omnidirectional spatial reasoning. The first stage measures a model’s ability to construct spatially grounded cognitive maps from panoramic images, whereas the second evaluates its capacity to solve reasoning tasks such as object counting, relative distance, and direction. We adopt Hungarian matching with rotation-invariant F1 scores for cognitive map alignment, and employ both rule-based and LLM-based evaluators to assess question-answering performance, including hallucination sensitivity. We further evaluate a suite of eight state-of-the-art MLLMs spanning both proprietary (*e.g.*, GPT-4o, Gemini 1.5 Pro) and open-source families (*e.g.*, InternVL2.5 [8], LLaVA-v1.5 [23], LLaMA3.2-Vision [11], Qwen2.5-VL [3], DeepSeek-VL2 [28], Janus-Pro [7]). All evaluations are conducted in strict zero-shot settings without any task-specific fine-tuning. To ensure methodological consistency and result reproducibility, we apply greedy decoding across all models.

In summary, our work makes the following contributions: **(I)** We introduce **OSR-Bench**, the first benchmark specifically tailored for evaluating the spatial reasoning abilities of Multimodal Large Language Models (MLLMs) in 360° panoramic environments. **(II)** We develop a scalable data generation pipeline that constructs over 153,000 high-quality QA pairs across three spatial reasoning categories, grounded in omni-cognitive maps derived from photorealistic indoor layouts. **(III)** We propose a novel *negative sampling strategy* that injects non-existent distractor objects into both prompts and QA templates, enabling robust evaluation of hallucination and visual grounding fidelity. **(IV)** We design a two-stage evaluation framework combining rotation-invariant cognitive map matching with both rule-based and LLM-based QA assessment, supporting interpretable and reproducible benchmarking. **(V)** We conduct a comprehensive evaluation across **8 state-of-the-art MLLMs**, both proprietary and open-source, under strict zero-shot settings, establishing the first comparative study of panoramic spatial reasoning in the multimodal era.

2. Related Work and Motivation

Visual Understanding and Reasoning. Visual understanding and reasoning lie at the intersection of computer vision and natural language processing, with tasks like Visual Question Answering (VQA) serving as key benchmarks. Datasets such as VQAv2 [12], GQA [13], and TextVQA [33] have driven progress in this area by testing increasingly com-

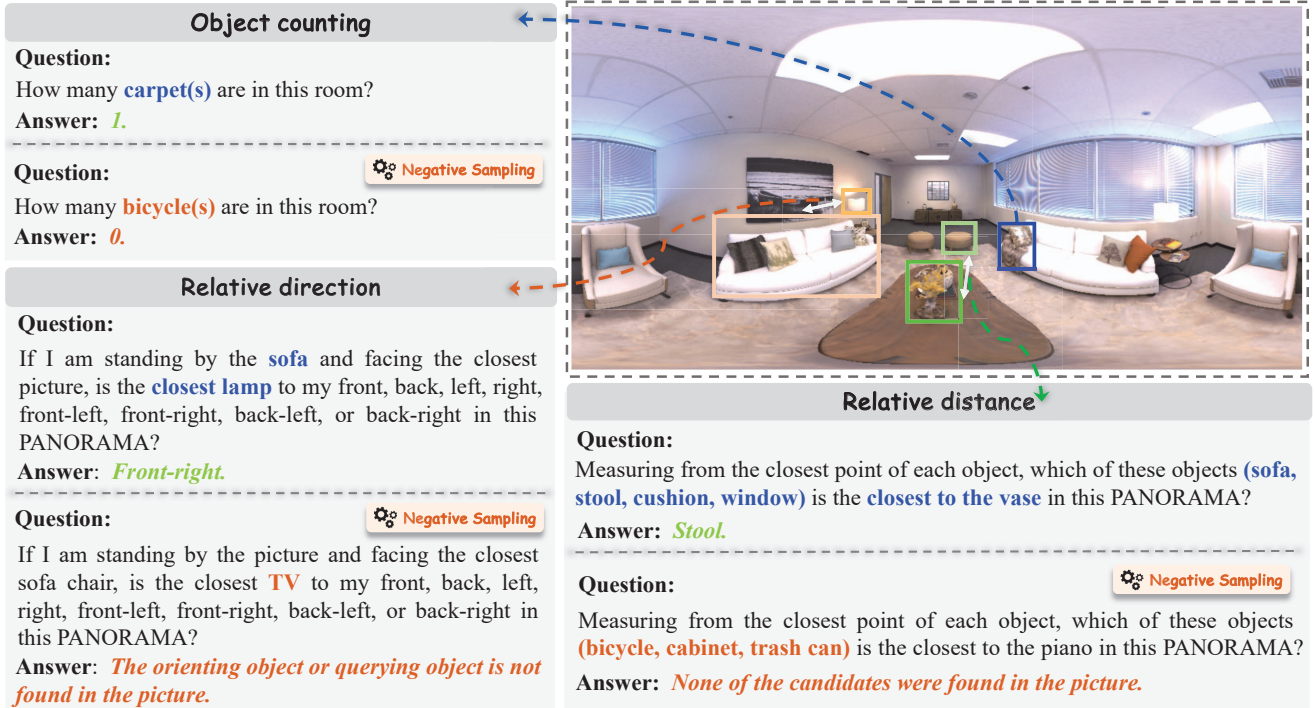


Figure 2. Cases of the established Omni-Spatial-Reasoning dataset.

plex reasoning skills. More recent benchmarks like SEED-Bench [17], MMBench [25], and MM-Vet [41] extend these efforts to multimodal large language models (MLLMs) across diverse visual-linguistic tasks. For multi-turn dialogue, benchmarks like ConvBench [24] and MMDU [26] further push the field by introducing hierarchical capabilities and multi-image contexts, respectively. However, challenges like object hallucination persist, as highlighted by POPE [20] and NOPE [27], which measure models’ tendency to generate false object references. Spatial reasoning has also received attention, with datasets like What’s-Up [35] and VSI-Bench [40] evaluating how effectively models handle spatial relationships and cognitive mapping.

Motivation. Despite these advancements, most existing benchmarks focus on standard field-of-view images, limiting their ability to assess spatial understanding in more complex environments. Notably, they overlook the unique challenges of omnidirectional perspectives, which involve handling object relationships, counting, and relative direction in panoramic scenes. To address this gap, we propose OSR-Bench, the first benchmark specifically designed to evaluate spatial reasoning in 360° environments, providing a more comprehensive assessment of MLLMs.

Panoramic Scene Understanding. Panoramic scene understanding presents unique challenges compared to conventional narrow-field vision. The equirectangular projection used for 360° images introduces geometric distortion, while the expanded field of view increases object density and spa-

tial complexity, complicating scene interpretation. Early work [10, 43, 48] in this area focused on specific tasks like room layout reconstruction, recovering instance-level geometry from panoramic images. However, these approaches often produce task-specific outputs (e.g., corner coordinates, mesh vertices) that are not directly compatible with Multimodal Large Language Models (MLLMs). Recent efforts have expanded to fine-grained scene understanding, including semantic segmentation [4, 44–47, 49, 51–53, 56, 57], aiming to capture richer semantic content. This growth has been accompanied by the creation of specialized panoramic datasets. For example, Pandora [38] focuses on orientation-aware object detection, while 360+x [6] and R3DS [37] provide multimodal annotations for comprehensive scene understanding. Large-scale outdoor datasets like KITTI-360 [22], DensePASS [30], and the Waymo Open Dataset [31] target autonomous driving, while Matterport3D [5] and Stanford2D3D [2] offer richly annotated indoor scans for embodied intelligence.

Motivation. Despite these advancements, existing datasets face several limitations when used to evaluate MLLMs under a unified VQA paradigm: ① *Task mismatch:* Many datasets focus on geometric reconstruction or dense prediction, which are not naturally suited for VQA-style evaluation. ② *Translation overhead:* Converting these tasks into natural language queries requires extensive post-processing, often losing geometric context. ③ *Limited spatial QA coverage:* Current omnidirectional VQA datasets (e.g., Pano-

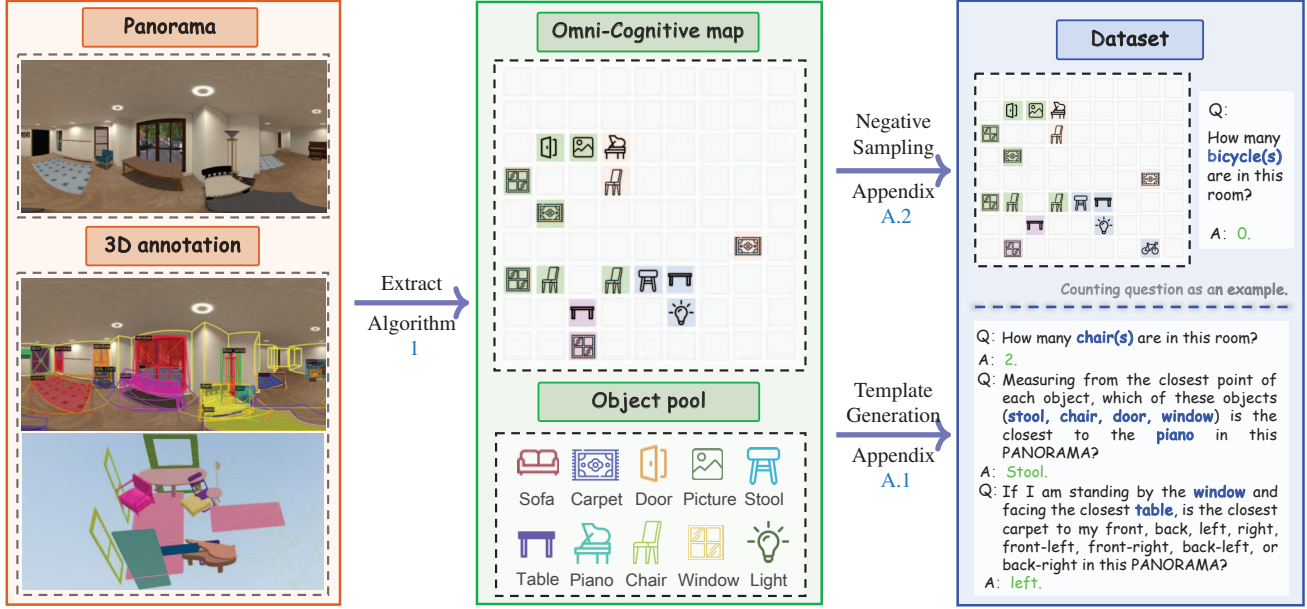


Figure 3. The data generation pipeline of our dataset.

AVQA [42], VQA 360° [9]) lack comprehensive spatial reasoning, particularly for object positions across the full 360° field of view. ④ *Handcrafted bias*: Manually constructed QA pairs may not capture the full complexity of panoramic spatial reasoning, increasing the risk of hallucination. To address these gaps, we propose **OSR-Bench**, a benchmark specifically designed to systematically assess spatial reasoning in panoramic scenes, providing a foundational and open resource for the community.

3. OSR-Dataset & Benchmark

To explore the spatial reasoning ability of existing MLLMs, we develop an omnidirectional question answering construction pipeline to effectively and automatically generate high-quality QA pairs at scale, as illustrated in Fig. 3. The following is the detailed dataset construction pipeline.

3.1. Dataset Construction

Data Collection. Our dataset is primarily built upon two public panoramic scene understanding datasets: ReplicaPano-Dataset [10] and the dataset introduced in DeepPanoContext [43]. Both datasets are derived from iGibson synthetic environments [18], providing photorealistic indoor panoramic scenes with comprehensive layout annotations. These datasets are originally designed for room layout reconstruction tasks. They contain detailed ground truth information about object positions, orientations, and room boundaries, enabling robust omni-cognitive map generation that is critical to our benchmark.

Omni-Cognitive Map Extraction. From the layout annotations in the source datasets, we extract ground truth omni-cognitive maps representing top-down views of each

room, as shown in Figure 3. These omni-cognitive maps serve as simplified representations of object locations within the environment, similar to the approach in [40]. These omni-cognitive maps serve two purposes: ① The omni-cognitive maps serve as **ground truth** references for evaluating the spatial understanding capabilities of LLMs that have been prompted to construct cognitive maps prior to addressing spatial reasoning questions, and ② The omni-cognitive maps enable **automatic generation** of diverse spatial reasoning questions with verifiable answers using predefined templates. More specifically, we begin with each panoramic image and its associated 3D layout annotations, from which we extract all object instances and construct a ground-truth omni-cognitive map representing the top-down layout of that scene. The same extracted objects are simultaneously organized into an object pool. Both the omni-cognitive maps and the object pools are then fed into our QA template generation module, where predefined templates are applied to produce diverse spatial-reasoning questions.

Negative Sampling. Unlike the existing spatial reasoning benchmark VSI-bench [40], which relies solely on ground-truth objects to construct QA templates and generate prompts for cognitive map, our approach addresses an important limitation: **models may prioritize knowledge from textual input while ignoring potential mismatches between visual and textual information**, as well as **hallucination issues** in visual information. Drawing inspiration from POPE [20], we implement negative sampling strategies (Popular Sampling and Adversarial Sampling, see Appendix A.2 for details) by introducing objects that do not exist in each scene. These interference items are incorporated into QA templates and prompt words that guide the model in cognitive map gen-

Algorithm 1 Omni-Cognitive Map Extraction

Require: Scene annotations S , Grid size $G = 10 \times 10$

- 1: Extract layout boundaries $(x_{min}, y_{min}, x_{max}, y_{max})$ from S
- 2: Initialize cognitive map C as $G \times G$ grid
- 3: Define mapping function: $(x, y) \mapsto (i, j)$ where $i \leftarrow \lfloor \frac{x-x_{min}}{x_{max}-x_{min}} \cdot G \rfloor, j \leftarrow \lfloor \frac{y-y_{min}}{y_{max}-y_{min}} \cdot G \rfloor$
- 4: **for** each object o with centroid (x, y) in S **do**
- 5: $(i, j) \leftarrow$ mapping of (x, y) to grid coordinates using the defined function
- 6: $C[i][j] \leftarrow C[i][j] \cup o.class$ //Add object class to grid cell
- 7: **end for**
- 8: $\mathcal{D} \leftarrow$ dictionary where $\mathcal{D}[c]$ is the count of class c in C
- 9: **return** $C, \mathcal{D}$

Table 1. **Comparison of OSR-bench with existing QA datasets.** **Modality:** multimedia input data type; **Auto:** whether the production process is automatic or manual craft **Dis:** presence of negative distractors by negative sampling to test hallucination; **#QAs:** number of QA pairs.

Dataset	Modality	Domain	Auto	Dis	#QAs
Pano-AVQA [42]	360° videos & audio	simple spatial & audio reasoning	✓	✓	51.7K
VQA 360° [9]	360° images	simple spatial & attributes reasoning	✓	✓	16K
VIEW-QA [34]	360° videos & audio	daily tasks of the visually impaired	✓	✓	4K
VSI-Bench [40]	NFOV videos	complex spatial reasoning	✓	✓	5K
OSR-Bench(ours)	360° images	complex spatial reasoning	✓	✓	153K

eration, resulting in two versions of our QA dataset. This methodology primarily evaluates a model’s visual-spatial understanding capabilities within a single-image panoramic perspective, focusing on three problem categories: object counting, relative distance, and relative direction, as illustrated in Figure 2.

Dataset Overview. Our *OSR-Bench* comprises 4,100 panoramic images with a total of 65 object categories sourced from 1,526 photorealistic synthetic indoor scenes. For each image, we generate 5 to 10 regular QA pairs for each evaluation category described in the previous paragraph, alongside an equal number of negative sampling QA pairs. To ensure data quality, we implement rigorous controls over the sampling process for both regular and negative sampling QA pairs, thereby eliminating ambiguity and preventing duplication in QA pair generation. In total, our comprehensive benchmark contains approximately 153k QA pairs, which lays a robust foundation for evaluating panoramic spatial reasoning capabilities compared to other existing related benchmarks in Table 1. Statistical insights are in Appendix B.

3.2. Our Evaluation Method

OSR-Bench can be viewed as a two-round visual dialogue task. Accordingly, we adopt a two-stage evaluation framework tailored to omnidirectional spatial reasoning. The first stage measures a model’s ability to construct accurate cognitive maps from panoramic images based on a comprehensive

prompt. The second stage evaluates the model’s capacity to answer specific spatial reasoning questions grounded in the constructed map. The overall evaluation pipeline is illustrated in Figure 4.

Metrics. For cognitive map evaluation, we utilize the Hungarian algorithm to determine the optimal assignment between predicted and ground truth object locations. Given a predicted cognitive map P and a ground truth cognitive map G , where P_i represents the predicted positions of object class i and G_i represents the ground truth positions, we define a distance matrix $D^i \in \mathbb{R}^{|G_i| \times |P_i|}$ where each element D_{jk}^i represents the Euclidean distance between the j -th ground truth position and the k -th predicted position for object class i :

$$D_{jk}^i = \sqrt{(G_j^i[0] - P_k^i[0])^2 + (G_j^i[1] - P_k^i[1])^2} \quad (1)$$

The Hungarian algorithm solves the assignment problem by finding the assignment matrix $X^i \in \{0, 1\}^{|G_i| \times |P_i|}$ that minimizes the total distance:

$$\min_{X^i} \sum_{j=1}^{|G_i|} \sum_{k=1}^{|P_i|} D_{jk}^i X_{jk}^i, \quad (2)$$

subject to the constraints:

$$\begin{aligned} \sum_{j=1}^{|G_i|} X_{jk}^i &\leq 1, \quad \forall k \in \{1, 2, \dots, |P_i|\}, \\ \sum_{k=1}^{|P_i|} X_{jk}^i &\leq 1, \quad \forall j \in \{1, 2, \dots, |G_i|\}. \end{aligned} \quad (3)$$

To accommodate potential rotational ambiguities in panoramic perception, we evaluate the cognitive map under rotational transformations and select the one that yields the highest F1 score. For each rotation r , we obtain a transformed prediction P^r and compute:

$$F1^r = \frac{2 \times \text{Precision}^r \times \text{Recall}^r}{\text{Precision}^r + \text{Recall}^r}, \quad (4)$$

where Precision^r and Recall^r are calculated based on matches between P^r and G using a distance threshold of 2.0 grid units. For hallucination assessment, we use CHAIR (Confident Hallucination Assessment in Image Reasoning) metrics such as POPE [20]: CHAIR_S, which indicates whether hallucination occurs in a scene(binary); CHAIR_I, which measures the proportion of hallucinated classes among predicted classes; and CHAIR_instance, which measures the proportion of hallucinated instances among all predicted instances. For regular spatial reasoning problems, we employ regex pattern matching to evaluate model responses against ground truth answers, which typically consist of concise words or phrases. Besides, we adopt category-specific evaluation metrics as follows: ① For **Object counting** tasks, we utilize a percentage-based accuracy measurement—awarding 100% when the model’s count exactly matches the ground truth, with scores decreasing proportion-

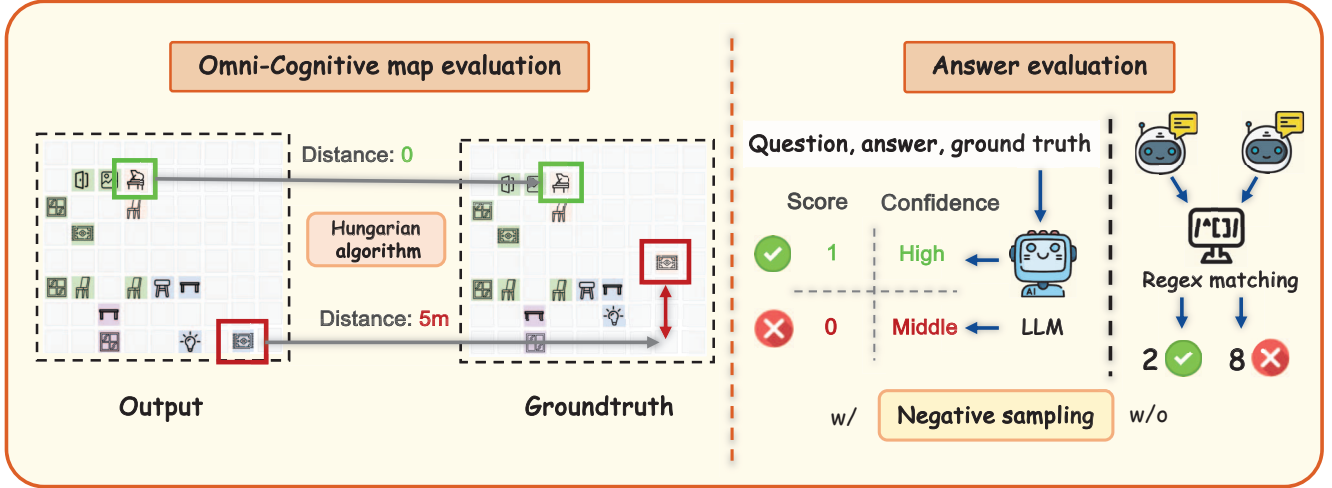


Figure 4. Benchmark Evaluation Pipeline.

ally as the numerical difference increases. ② For *Relative distance* and *Relative direction* tasks, we implement binary scoring (correct/incorrect) to assess response accuracy, given the discrete nature of these spatial relationships.

LLM Evaluator For more complex evaluation scenarios, particularly with negative sampling, we adopt an LLM-based evaluator—which has been widely used in the evaluation of open-ended questions [50], due to the challenge of evaluating answers about non-existent objects. The LLM evaluator provides binary scores (0 or 1) and confidence levels (high, medium, low), providing detailed evaluation for ambiguous cases. Compared with traditional rule-based evaluation methods, it demonstrates robustness to linguistic variations in model outputs and maintains evaluation fairness across different models; compared with manual assessment methods, this method is more cost-effective and achieves greater objectivity and reduces bias. The specific prompts used for the LLM evaluator to evaluate each spatial reasoning task are detailed in the Appendix C.3.

3.3. Benchmark Experiments

We evaluate 8 MLLMs across diverse series. Examples of proprietary models include Gemini-1.5-Pro [32] and GPT-4o [1]. For open-source models, we evaluate models from InternVL2_5-78B [8], LLaVA-v1.5-13B [23], Llama-3.2-90B-Vision-Instruct [11], Qwen2.5-VL-72B-Instruct [3], deepseek-vl2 [28] and Janus-Pro-7B [7]. To ensure methodological rigor and comparative validity, all evaluations are conducted under zero-shot settings without any task-specific fine-tuning. Furthermore, we employ greedy decoding across all models to maximize the reproducibility of experimental results. Appendix D details the setup, while Appendix E reports additional results on newer post-submission MLLMs.

Table 2. **Omni-Cognitive Map Generation Performance (with Success Rate of different models in constructing properly formatted cognitive maps)**. Model performance on generating omni-cognitive maps from benchmark images with identical prompts, without negative sampling. **Best** results are in bold, and second-best are underlined.

Model	Avg dist.↓	Precision	Recall	F1	Succ. Rate(%)
GPT-4o [1]	<u>3.501</u>	0.333	<u>0.219</u>	<u>0.259</u>	<u>97.50</u>
Gemini-1.5-Pro [32]	3.475	<u>0.313</u>	0.238	0.267	99.50
Llama-3.2-90B [11]	5.991	0.256	0.188	0.212	79.15
Qwen2.5-VL-72B [3]	4.071	0.307	0.199	0.231	95.60
InternVL2_5-78B [8]	5.436	0.291	0.218	0.243	92.40
LLaVA-v1.5-13B [23]	8.512	0.184	0.153	0.167	10.84
deepseek-vl2 [28]	9.607	0.133	0.146	0.124	44.83
Janus-Pro-7B [7]	14.467	0.111	0.131	0.120	4.29

3.3.1. Results

Cognitive Map Generation Capabilities. As in Table 2, proprietary models generally outperform open-source alternatives in generating omni-cognitive maps from panoramic images. Gemini-1.5-Pro achieves the highest F1 score of 0.267, followed by GPT-4o at 0.259. Among open-source models, InternVL2-5-78B performs best with an F1 score of 0.243, approaching the proprietary models’ results. However, there is a substantial gap between model families, with the lowest-performing model (Janus-Pro-7B) scoring only 0.008. Notably, all models exhibit poor performance, with the best precision and recall values reaching only 0.333 and 0.238, respectively, highlighting the challenges in accurately capturing spatial arrangements in omnidirectional contexts.

Impact of Cognitive Maps on Spatial Reasoning. Table 3 reveals interesting patterns regarding the utility of cognitive maps for spatial reasoning tasks. For object counting, all models benefit from having a cognitive map, with improvements ranging from 2% to 15%. However, the impact

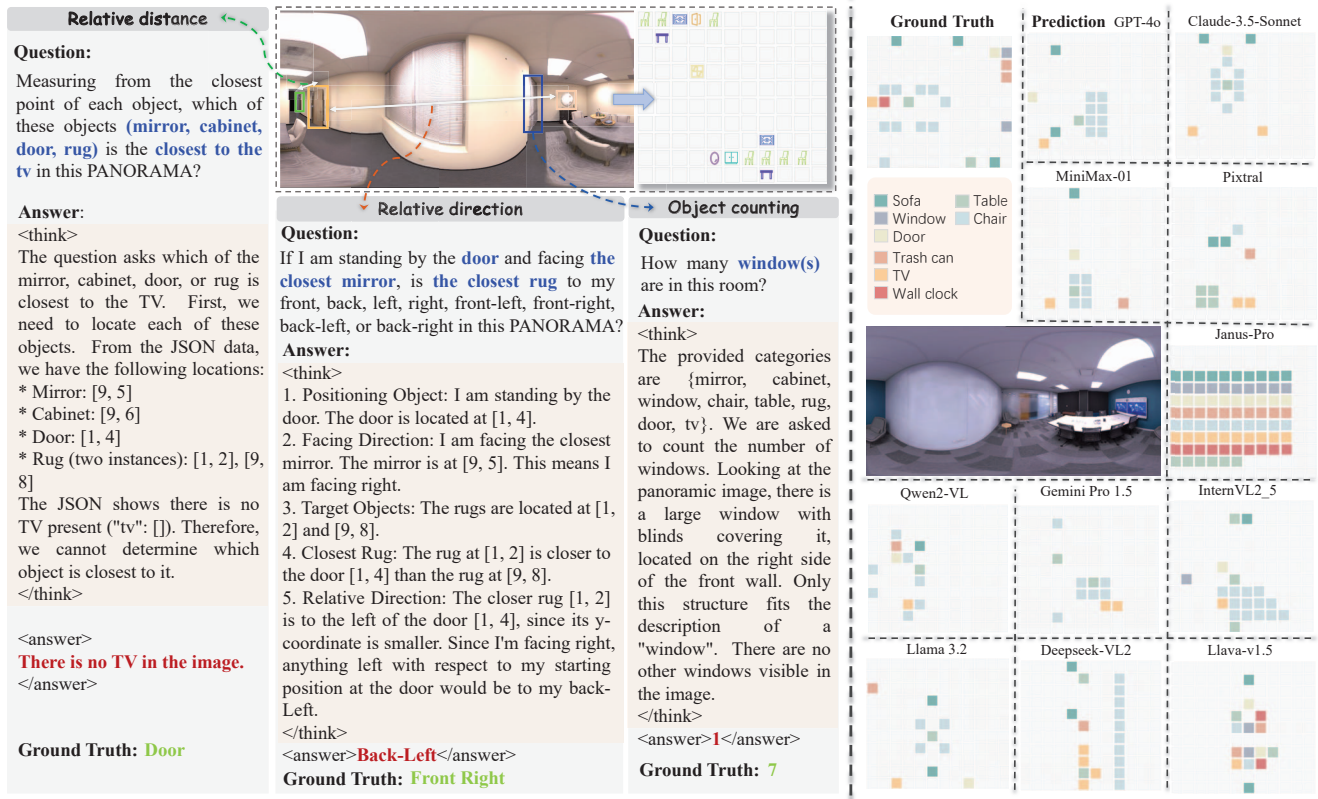


Figure 5. Qualitative examples of omnidirectional spatial reasoning on our OSR-dataset.

Table 3. Comparison of the average scores of the models for each type of question without negative sampling. “w/ cogmap” indicates models were first prompted to generate a cognitive map of the omnidirectional scene prior to answering spatial questions, while “w/o cogmap” indicates models answered questions directly without the cognitive mapping step.

Model	object count		relative distance		relative direction	
	w/ cogmap	w/o cogmap	w/ cogmap	w/o cogmap	w/ cogmap	w/o cogmap
GPT-4o [1]	0.696	0.645	0.342	0.294	0.216	0.285
Gemini-1.5-Pro [32]	0.744	0.711	0.417	0.424	0.074	0.109
Llama-3.2-90B [11]	0.697	0.674	0.368	0.323	0.196	0.123
Qwen2.5-VL-72B [3]	0.558	0.498	0.299	0.325	0.156	0.181
InternVL2_5-78B [8]	0.683	0.616	0.365	0.397	0.059	0.087
LLaVA-v1.5-13B [23]	0.598	0.553	0.245	0.226	0.144	0.169
deepseek-v12 [28]	0.734	0.579	0.091	0.055	0.129	0.131
Janus-Pro-7B [7]	0.487	0.405	0.200	0.159	0.068	0.159

on relative distance reasoning is mixed: some models (notably GPT-4o, Llama-3.2-90B, and deepseek-v12) improve substantially with cognitive maps, while others (Gemini-1.5-Pro, Qwen2.5-VL-72B) perform slightly better without them. Most surprisingly, for relative direction tasks, seven out of eight models perform better without cognitive maps, with only Llama-3.2-90B showing improvement when using cognitive maps. This counterintuitive finding suggests that the imperfect cognitive maps generated by these models may actually interfere with their innate ability to reason about directional relationships in panoramic images. The cognitive

overhead of reconciling potentially inaccurate map representations with the original visual information appears to degrade performance on orientation-sensitive tasks, while still providing benefits for simpler counting operations. The qualitative examples of omnidirectional spatial reasoning are shown in Figure 5 and more case are shown in Appendix F.

Hallucination Robustness. Under the negative sampling condition (Table 4), all models show substantial hallucination tendencies, with CHAIR_S scores ranging from 0.767 to 0.884 (lower is better). Proprietary models are more robust, with GPT-4o achieving the lowest CHAIR_I (0.198) and CHAIR_instance (0.180) scores, indicating fewer hallucinated object classes and instances. Among open-source models, Qwen2.5-VL-72B shows strong resistance, with the lowest CHAIR_S score of 0.767. All models experience performance degradation in spatial reasoning tasks when negative distractor objects are introduced, highlighting limited robustness to interference. Interestingly, hallucination resistance does not perfectly correlate with spatial reasoning performance—deepseek-v12 achieves the highest object counting score (0.641) despite relatively high hallucination metrics. This discrepancy likely results from deepseek-v12’s architecture, which includes specialized tokens for object localization and reasoning, allowing strong counting capa-

Table 4. **Model Performance with Negative Sampling.**

Model	Precision	Recall	CHAIR_S	CHAIR_I	CHAIR_ins	obj count	rel distance	rel direction
GPT-4o [1]	0.303	0.197	0.791	0.198	0.180	0.589	<u>0.435</u>	0.137
Gemini-1.5-Pro [32]	0.269	0.211	0.852	0.202	0.180	0.602	0.441	<u>0.230</u>
Llama-3.2-90B [11]	0.238	0.201	0.875	0.339	0.339	0.535	0.414	0.140
Qwen2.5-VL-72B [3]	0.303	0.189	0.767	0.314	0.306	0.579	0.319	0.232
InternVL2_5-78B [8]	<u>0.278</u>	<u>0.210</u>	<u>0.782</u>	0.248	<u>0.243</u>	0.567	0.378	0.156
LLaVA-v1.5-13B [23]	0.083	0.062	0.884	0.291	0.284	0.523	0.175	0.103
deepseek-vl2 [28]	0.095	0.124	0.844	0.380	0.373	0.641	0.064	0.047
Janus-Pro-7B [7]	0.001	0.002	0.872	0.363	0.277	0.510	0.160	0.004

Table 5. **Thinking-mode Prompt Result.** Performance improvements for Gemini-1.5-Pro using thinking-mode prompt (with and without Omni-cognitive map) on the non-negatively sampling version of the dataset.

	object count	rel distance	rel direction
w/ omni-cogmap	0.745 (+0.001)	0.424 (+0.007)	0.135 (+0.061)
w/o omni-cogmap	0.748 (+0.037)	0.434 (+0.009)	0.092 (−0.018)

bilities despite weaker hallucination resistance.

Instruction Following and Format Adherence The vast differences in instruction-following capabilities across models are quantitatively demonstrated in Table 2, which shows the success rates in generating properly formatted cognitive maps. Proprietary models excel at this task, with Gemini-1.5-Pro and GPT-4o achieving success rates of 99.50% and 97.50% respectively. Among open-source models, larger models like Qwen2.5-VL-72B (95.60%) and InternVL2_5-78B (92.40%) approach proprietary-level performance, while smaller models struggle dramatically. The success rate drops to just 10.84% for LLaVA-v1.5-13B and a mere 4.29% for Janus-Pro-7B. This strong correlation between parameter count and instruction-following capability explains much of the performance gap observed in Table 2, which shows that models with low success rates in generating correctly formatted cognitive maps also exhibit poor F1 scores, reflecting both limited visual-spatial understanding and difficulty following structured response protocols for complex spatial tasks.

Enhanced Reasoning Through CoT Prompt. The results in Table 5 show that prompting models to use structured reasoning improves performance across most tasks. For example, Gemini-1.5-Pro gains +0.037 in object counting without cognitive maps when guided to think step-by-step (templates in Appendix C.2). This suggests that explicit reasoning scaffolds are particularly helpful when models rely solely on internal scene representations. However, gains are generally smaller when cognitive maps are provided, indicating that external spatial representations may overlap with structured reasoning. An exception is relative direction reasoning, which improves significantly (+0.061) with thinking-mode prompts, suggesting that explicit reasoning helps models better integrate directional information from cognitive maps. Overall, structured thinking appears to en-

hance the effective use of cognitive maps, reducing potential confusion and improving spatial understanding. The qualitative results in Figure 5 confirm these findings.

4. Conclusion

In this paper, we introduced **OSR-Dataset** and **OSR-Bench**, the first benchmarks specifically designed to evaluate the spatial reasoning capabilities of Multimodal Large Language Models (MLLMs) in 360° panoramic environments. Our benchmark includes over 153,000 question-answer pairs across three core spatial reasoning tasks—object counting, relative distance, and relative direction—grounded in high-fidelity 3D scene layouts. To improve evaluation rigor, we proposed a negative sampling strategy to systematically test for hallucination and grounding robustness. We also developed a two-stage evaluation framework that combines rotation-invariant cognitive map alignment with both rule-based and LLM-based question answering, providing fine-grained, interpretable metrics. Our evaluation of eight state-of-the-art MLLMs revealed significant gaps in their spatial reasoning performance under panoramic settings, underscoring the need for more perceptually grounded architectures.

Discussion While OSR-Bench provides a comprehensive framework for evaluating omnidirectional spatial reasoning, several limitations remain. The benchmark primarily focuses on indoor environments and may not capture outdoor complexities such as dynamic objects, varying weather, and large-scale spatial structures. Our work advances perceptually grounded MLLMs, with applications in autonomous navigation, augmented reality, and robotics. However, improved spatial reasoning may raise privacy and surveillance concerns, as panoramic cameras are widely used in security systems. There is also a risk of misuse in military or surveillance contexts. Researchers should consider these ethical implications when deploying MLLMs trained on spatial reasoning benchmarks.

Our future work will focus on: (I) extending to real-world scenes beyond synthetic indoor environments; (II) evaluating generalization to out-of-distribution scenarios, including unfamiliar objects and novel scenes; (III) developing more sophisticated reasoning tasks for dynamic scene understanding; and (IV) exploring post-training strategies to enhance panoramic spatial reasoning.

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